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(1) Research Program/Project/Study Title: Selected fermented plant-based protein meal as replacement to SOYBEAN MEAL in the growth and survival of juvenile Nile Tilapia and Catfish	
Proponent Name:	Robert Christian F. Patani
Academic Rank:	Instructor 1
Co-Authors (with academic rank):	Dr. Rey J. dela Calzada, Assoc Prof IV.; Ianthe Marie P. Benliro, Assoc Prof III; Gelene Roda I. Alvarez, Instructor 1; Alessandra Nicole Falalimpa, Contractual Faculty; Noreen Beah M. Alar; SRS 1; Kate Alyssa G. Jore; SRS 1
Campus/Satellite College/Address:	Capiz State University – Pontevedra, Bailan, Pontevedra, Capiz
Telephone/Cell Phone/Fax Number:	(+63) 950 937 2319
E-mail Address:	rcfpatani@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Pontevedra
		Province:	Capiz
		Region:	Western Visayas
(5) Classification Research: Development: ☐ Basic ☒ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Tilapia; Pantat; Mung Beans/Mungo; Stringbeans/Sitaw; Madre de Agua or <i>Trichanthera gigantea</i>; True Indigo or <i>Indigofera tinctoria</i>; Water Fern or <i>Azolla pinnata</i>		(8) Discipline/Field of Specialization Aquaculture; Fish Nutrition	

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Aquaculture in the Philippines, driven by the need for food security, relies heavily on fishmeal as a primary animal-based protein source in aquafeed. However, challenges such as high costs and limited availability have prompted exploration of cheaper plant-based alternatives (Dernekbası & Karayucel, 2017). Soybean meal (SBM) is a preferred option due to its high protein content and amino acid profile, but its expense and dependence on imports pose economic drawbacks (Hajra et al., 2013; Ogello et al., 2014; Gaxiola et al., 2017).

Recent studies have investigated various plant-based alternatives to SBM, including mung beans, string beans, madre de agua leaf, indigo leaf, and even aquatic plants like *Azolla*, aiming to reduce reliance on fishmeal and SBM. Mung beans, known for affordability and rich nutritional content, have successfully replaced traditional protein sources in aquaculture diets, enhancing growth performances and nutrient utilization in various species (de Silva & Gunasekera, 1989; Ganzon-Naret, 2014a; Ganzon-Naret, 2014b; Apines-Amar et al., 2015; Soliman et al., 2018). String beans, a widely cultivated and inexpensive crop in the Philippines, have been explored as a protein source in aquafeed. Studies indicate that protein concentrates from string beans can replace fish meal without adverse effects on growth parameters and intestinal tissue in species such as Nile Tilapia, Asian Seabass, Giant Freshwater Prawn, and Pacific White Shrimp (Eusebio & Coloso, 2000; Rivas-Vega et al., 2006; Lara-Flores et al., 2007; Olivera-Castillo et al., 2011; Aya et al., 2014). Likewise, Madre de Agua (*Trichanthera gigantea*) has been utilized as livestock and poultry feed ingredient and an alternative protein source with 18-22% crude protein without affecting their growth performance (Rosales, 1997; Ly et al., 2001; Libatique, 2020). Moreover, *Indigofera tinctoria* or Philippine Indigo has also served as an alternative protein source for ruminants, poultry, and even in various fish species such as juvenile carp and gourami fingerling that demonstrated improved growth performance, protein retention, and feed efficiency (Ginting et al., 2010; Faradillah et al., 2015; Setiyowati et al., 2021; Rakhmawati et al., 2023). Lastly, *Azolla*, with a promising crude protein profile of 20-35% and rich amino acid counts that surpasses soybean meal in most cases, has also been investigated and successfully employed as an alternative protein source for various fish species, contributing to improved growth, health, feed utilization, and survival rates (Goda et al., 2018; Magouz et al., 2020; Mosha et al., 2020; Noviyanti et al., 2020).

Despite favorable findings, plant-based protein sources face challenges due to high crude fiber content, anti-nutritional factors and inhibitors impacting the growth of the species and the apparent digestibility coefficient, leading to increased organic matter discharge (Allan et al., 2020; Vo et al., 2020; Gaxiola et al., 2017). Techniques such as heating, extrusion, germination, and fermentation are explored to address these issues. Likewise, fermentation using microorganisms like *Bacillus* spp., *Aspergillus* spp., *Lactobacillus* spp., and *Saccharomyces* spp., shows promise in breaking down complex molecules and improving protein quality (Yang et al., 2016). The application of fermentation on various plant-based protein sources such as soybean and cottonseed meal and their applicability as alternative protein source have already been explored (Barnes et al., 2012; Shiu et al., 2013; Sun et al., 2014; Sun et al., 2015; Li et al., 2020). However, the utilization of other viable and fermented plant-based protein sources as alternative protein sources for the diets of valuable cultured fish species are yet to be explored.

Hence, this study explores the potentiality of various fermented plant-based protein sources such as mungo meal, sitaw meal, *Trichanthera* leaf meal, *Indigo* meal, and *Azolla* meal as replacement to Soybean Meal in the diets of Nile Tilapia and African Catfish. Results of this study will serve as viable information for formulating diets of cultured fish species by utilizing locally available and affordable plant materials, thereby reducing cost and reliance to soybean meal, and eventually to fishmeal, and achieving food security.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

This project addresses the issue of high aquafeed costs due to the utilization of costly fishmeal and

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soybean meal as the primary sources of protein for the fish diets. Five (5) plant-based materials that are known to be a viable source of protein namely, mungo beans, sitaw beans, madre de agua, Philippine Indigo, and *Azolla* will be investigated for their potential as replacement to soybean meal. Fermentation of mungo, sitaw, Madre de Agua, Philippine Indigo, and *Azolla* using isolated *Bacillus subtilis* from adult Nile Tilapia, will be the biotechnological method to be employed to reduce crude fiber content, improve the digestibility, and remove antinutritional factors present in the aforementioned plant materials. For the first phase of this project, the various isoproteic and isoenergetic formulated diets using Fermented Mungo Meal (FMM), Fermented Sitaw Meal (FSM), Fermented *Trichanthera* Leaf Meal (FTLM), Fermented *Indigofera* Meal (FIM) and Fermented *Azolla* Meal (FAM) will be incorporated to the diets of Juvenile Nile Tilapia; meanwhile the second phase of the project will revolve on the incorporation to the diets of Juvenile African Catfish. The apparent digestibility of the formulated diets will be determined through an indirect method using an inert marker mixed with the formulated diets. Nutrient composition of the diet will be analyzed through a third-party analysis. Growth performance, feed efficiency, and survival rate of the species will also be analyzed through the 45-day feeding trial period.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Generally, this project proposal aims to determine the potentiality and effects of replacing soybean meal protein with selected fermented plant-based protein diets in Juvenile Nile Tilapia (*Oreochromis niloticus*) and Juvenile African Catfish (*Clarias gariepinus*).

Specifically, this project proposal wants to:

- Determine and compare the apparent digestibility coefficient, nutrient composition, and inhibited antinutritional factors of the selected fermented plant-based protein diets: Fermented Mungo Meal (FMM), Fermented Sitaw Meal (FSM), Fermented *Trichanthera* Leaf Meal (FTLM), Fermented *Indigofera* Meal (FIM) and Fermented *Azolla* Meal (FAM); and,
- Evaluate the growth performance, feed intake, feed utilization, and survival of juvenile Nile Tilapia and juvenile African Catfish fed with selected fermented plant-based protein diets at various inclusions.

(11) Review of Literature/Prior Art Search (Related researches/activities that have been conducted. State-of-the-art of current technology/information from which the project proposal will take-off).

Fishmeal has been the major animal-based protein source utilized in the aquafeed industry because of its high protein content, well-balanced amino acid profiles, and palatability. However, its utilization as the primary feed ingredient is faced with problems concerning the constant availability and high cost of raw materials, and other operational costs brought about by transporting and processing the ingredient (Dernekbası & Karayucel, 2017). Cheaper alternatives, specifically plant-based sources, were considered to replace the traditional feed ingredient without reducing the nutritional quality of the feedstuff. Soybean meal (SBM) is one of the most preferred plant-based protein sources because of its high protein content, better amino acid profile, and overall best protein quality amongst all plant-based sources that is at par with fish meal (Ogello et al., 2014). However, just like fishmeal, SBM is also expensive, and the usage of such ingredient in the Philippine context pose an economical downside, primarily because of high costs due to importation.

Hence, efforts to look for other cheaper and highly available plant-based protein sources for fish diets are essential. In present studies, other plant-based alternatives to SBM range from other legumes to oilseeds, brans, root tubers, and other sorts of plants, even aquatic macrophyte, are being investigated to reduce the dependence on fishmeal and SBM (Hajra et al., 2013; Ogello et al., 2014; Gaxiola et al., 2017). Amongst these wide range of plant-based alternatives, Mung beans or Munggo (*Vigna radiata*), Stringbeans or Sitaw (*Vigna unguiculata sesquipedalis*), Madre de Agua or (*Trichanthera gigantea*), True Indigo (*Indigofera tinctoria*), and Water Fern (*Azolla pinnata*) are being investigated on their capability to replace SBM.

Mung Beans or munggo, is also known to be one of the cheapest sources of plant protein (23-28%), at the same time minerals such as calcium, iron, magnesium, sodium, Vitamins A, B, C & E (DA-BPI, n.d.). In the country it has a total production of 36,849.05 MT for the year 2022 and Region 6 is among the top producer of such commodity (PSA, 2023). Mung bean has been used as an alternative source of protein for several aquaculture species such as Juvenile Milkfish (*Chanos chanos*, Forsskal), Sea

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Bass fingerlings, and Nile Tilapia fingerlings and fry (de Silva & Gunasekera, 1989; Ganzon-Naret, 2014a; Ganzon-Naret, 2014b; Apines-Amar et al., 2015; Soliman et al., 2018), exhibiting better growth performances of the species, nutrient utilization, and without posing adverse effects on the protein efficiency ratio, protein retention, hepatosomatic index (HIS), and liver and intestinal histology of the aforementioned fish commodities.

Moreover, Sitaw is an important crop in the Philippines since it widely grown even in home gardens as companion crops or commercial crops. For the year 2022, it had a total production of 107,893.47MT, where Western Visayas was also the top producing region of such commodity (PSA, 2023). Sitaw is also known to be a cheap yet rich source of protein (23-26%), Vitamins A, C, thiamin, riboflavin, iron, phosphorus, potassium, and magnesium (DA-ATI, n.d.). The utilization of protein concentrates from sitaw as substitution to fish meal in the diets of Nile Tilapia fry was initiated by Olvera-Novoa et al. (1997). Succeeding investigations further investigated the potential of sitaw in other aquaculture commodities such as the grow-out Nile Tilapia, Asian Seabass, and even in crustaceans such as Giant Freshwater Prawn (*Macrobrachium rosenbergii*), Pacific White Shrimp (*Litopenaeus vannamei*) which suggested that such crop can be a replacement protein source without affecting the growth parameters and intestinal tissue (Eusebio & Coloso, 2000; Rivas-Vega et al., 2006; Lara-Flores et al., 2007; Olivera-Castillo et al., 2011; Aya et al., 2014).

Furthermore, Madre de Agua (*Trichanthera gigantea*) is a fodder tree that is native to Colombia, but was later introduced to the Philippines, Vietnam, Cambodia, and other Latin-American countries (Cook et al., 2005). It has gained recognition among hog growers and poultry raisers in the Philippines for its valuable contribution to livestock growth; even before its introduction to the Philippines, feed manufacturers were already aware of its nutrient levels, recognizing it as a potential protein source with a content ranging from 18-22% in dry matter form. Generally, it has been used as a protein source for pigs, poultry, rabbits, goats, guinea pigs (Nhan et al., 1997; Ly et al., 2001; Kihn et al., 2014; Balraj et al., 2018; Libatique 2020). Notably, this protein is predominantly true protein, exhibiting a favorable amino acid balance (Rosales 1997).

Indigofera tinctoria, also known as true indigo or Philippine Indigo, is a leguminous fodder plant species characterized by its compound pinnate leaves, featuring small leaflets. It is indigenous to the tropical and subtropical regions, where cultivation in the Philippines was widely practiced (Garrity et al., 1994). This species has been profiled to have high nutritive value indicated by its high crude protein content (22-32%), high organic matter digestibility, and sufficient mineral content (Ca, Mg, Zn, and Mn) (Ginting et al., 2010). Hence, this plant has been utilized as an alternative protein source for ruminants, poultry, and poultry eggs (Faradillah et al., 2015; Gunun et al., 2023) without impairing the quality of the eggs, growth of the organism cultured. Likewise, the species has been investigated on its potential as feed ingredient and alternative protein source in the diets of fish such as gourami fingerling, juvenile carp, Hoven's Carp (*Leptobarbus hoevenii*), Striped Catfish (*Pangasianodon hypophthalmus*) (Suprayudi et al., 2012; Jefry 2021; Setiyowati et al., 2021; Pamukas et al., 2023; Rakhmawati et al., 2023), indicating that its inclusion in the diet resulted to better growth performance, protein retention, and feed efficiency.

Lastly, *Azolla* is a small aquatic fern and leafed floating plants that is native to the tropics, subtropics, and warm temperate areas in Asia, Africa, and America. This plant species is known for its rapid growth and wide array of practical usage such as green manure, insect repellent, herbicide, water purifier, nitrogen fertilizer, and even potential protein source for the aquafeed industry with its promising crude protein profile of 20-35% (Mosha, 2018; Yohana et al., 2023). The plant material is also rich in Essential Amino Acids, which are relatively higher than soybean meal (Mosha et al., 2020), except for histidine (Brouwer et al., 2018). *Azolla* meal has been used as an alternative protein source for various fish species including Tilapia species, Catfish, Carp, Rohu (*Labeo rohita*), Catla (*Catla catla*), and even in freshwater prawns (Goda et al., 2018; Magouz et al., 2020; Mosha et al., 2020; Noviyanti et al., 2020) which have exhibited improved growth performances, health conditions, feed utilization, and survival rate of the species.

While these findings were favorable, plant-based protein sources are deemed to be inferior protein sources because of the presence of some barriers, specifically the high crude fiber content (Ismail et al., 2022) and anti-nutritional factors and inhibitors that highly affects the growth of the cultured species, low apparent digestibility coefficient (Allan et al., 2020) and increase discharge of organic matter in the environment (Vo et al., 2020). To address the issue of the presence of high crude fiber content, anti-nutritional factors and inhibitors, heating or thermal processing and extrusion are widely acceptable

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techniques (Gaxiola et al., 2017), yet can have risks to reduce the protein quality of the material due to maillard reaction, over-cooking and underprocessing (Zhang and Parsons, 1996; Dozier and Hess, 2010) and loss of vitamins (Vo et al., 2020).

Biotechnological methods were also employed such as germination and fermentation, which are ecologically friendly and economically cheaper than the conventional and previous methods mentioned (Gaxiola et al., 2017). Specifically, fermentation involves the transformation of complex molecules to simpler one with the aid of microorganisms such as *Bacillus* spp., *Aspergillus* spp., *Lactobacillus* spp, and *Saccharomyces* spp., resulting to improved protein quality of plant-based protein ingredients by reducing peptide sizes of the protein molecules (Gao et al., 2013; Yang et al., 2016). At present, the application of fermented plant-based protein for aquafeed are only limited on studies using fermented soybean meal (Barnes et al., 2012; Shiu et al., 2013); fermented cottonseed meal (Sun et al., 2014; Sun et al., 2015), and several studies using fermented *Indigofera* on common carp (Suprayudi et al., 2012), fermented *Azolla* on Freshwater Bawal Fish (Ni'mah et al., 2023) and Thai Catfish (Afrin et al., 2023). However, investigations on the potential of fermented mungo meal, fermented sitaw meal, fermented *Trichanthera* leaf meal, fermented *Indigofera* meal, and fermented *Azolla* meal on fish species such as Nile Tilapia and African Catfish are still to be discovered.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories, and variables that indicate relevant relationships to be tested.)

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

This project will be divided into two (2) phases, where the first phase will focus on the Juvenile Nile Tilapia (*Oreochromis niloticus*) as the species of interest, and the second phase that will focus on the Juvenile African Catfish (*Clarias gariepinus*) as the species of interest; hence, the methodology presented will focus on the first commodity of interest. Subsequently, the experiment for each phase will undergo three (3) parts, namely: (1) fermentation of the selected plant-based protein sources; (2) digestibility performance analysis, and (3) growth and survival performance analysis of the species of interest.

Fermentation of Plant-based protein sources

The bacterial strain to be used for the fermentation of the different selected plant-based protein will be the *Bacillus subtilis* strain that will be isolated from the gut of adult Nile Tilapia, following the procedure of Efendi and Yusra (2014). The homogenate will be serially diluted to obtain a bacterial population of 10^{-8} with glucose tryptone agar as medium. Plates will then be incubated for 48 h at 37°C. For the fermentation part, 18 kgs each of munggo and threshed sitaw will be subjected to oil extraction by steaming at 90°C for 2 hours before grinding them to 50 µm as described by Vo et al. (2020). After grinding, the ground mungoo and sitaw beans, along with the *Tricanthera* leaf, *Indigofera* leaf, and *Azolla* will be dried at 80°C in a hot-air oven, grounded, and sieved (400 µm in diameter). Next, it will be moistened with water at 45% moisture content and steamed at 100°C for 30 minutes. *B. subtilis* will be inoculated to the Munggo Meal, Sitaw Meal, *Trichanthera* leaf meal, *Indigofera* Meal, and *Azolla* Meal and incubated at 37°C for 48 h (Chi & Cho 2015; Zhang et al. 2014). The FMM, FSM, FTLM, FIM, and FAM will then be dried in a hot-air oven at 60°C for 24 h. Afterwards, it will be finely ground and analyzed for its nutritional value.

Digestibility Performance Analysis

The ADC analysis of the selected fermented plant-based protein diets will be performed by an indirect method that includes 0.1% of Chromic Oxide (Cr_2O_3) as inert marker to the diets. A reference diet (RD) that will match the nutrient requirements of the species of interest and with SBM as the dietary protein source, as well as a five (5) test diet (TD) obtained by replacing 50% of SBM with FMM, FSM, FTLM, FIM, and FAM, respectively, as a protein source will be formulated to examine the digestibility of the diets. Juvenile Nile Tilapia will be stocked into twelve (12) 120-L rectangular plastic tanks (10 juveniles/tanks), where diets (RD, TD1 (FMM), TD2 (FSM), TD3 (FTLM), TD4 (FIM), and TD5 (FAM)) will be randomly assigned to tanks (n=3). The fish sample will first be acclimatized and fed for 7 days prior to fecal collection. This procedure will follow the method of da Silva et al. (2017) where siphoning is performed followed by pipetting. It will be done daily in the morning until a representative volume of feces is collected. Feces will then be oven-dried (50°C, 24 h), powdered and stored for posterior

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analysis.

Apparent digestibility coefficients will be calculated using the following formulae (da Silva et al. 2017):

$$\text{Apparent digestibility coefficients of a nutrient } ADC_n = 1 - \left[\left(\frac{\%Cr_2O_3 \text{ diet}}{\%Cr_2O_3 \text{ feces}} \right) \left(\frac{\%N \text{ feces}}{\%N \text{ diet}} \right) \right]$$

where, ADC_n = the apparent digestibility coefficients of a nutrient, $\%Cr_2O_3 \text{ diet}$ = % chromic oxide of the diet, $\%Cr_2O_3 \text{ feces}$ = % chromic oxide of the feces, $\%N \text{ feces}$ = % of nutrients (or kJ g⁻¹ gross energy) in feces and $\%N \text{ diet}$ = % of nutrients (or kJ g⁻¹ gross energy) in the test diet. The ADC of the different test ingredients will also be calculated using:

$$ADC_{\text{test ingredient}} = ADC_{\text{test diet}} + \left[(ADC_{\text{test diet}} - ADC_{\text{reference diet}}) \left(\frac{0.5 \times D_{\text{reference}}}{0.5 \times D_{\text{ingredient}}} \right) \right]$$

where, $D_{\text{reference}}$ = % of nutrients (or kJ g⁻¹ gross energy) in the reference diet mash (as is) and $D_{\text{ingredient}}$ = % of nutrients (or kJ g⁻¹ gross energy) in the test ingredient (as is).

Growth and Survival Performance Analysis

Sixteen (16) isoproteic (32% digestible protein for Nile Tilapia and 39% digestible protein for African Catfish) and isoenergetic (17.6 kJ/g digestible energy for Nile Tilapia and 11.7% kJ/g digestible energy for African Catfish) diets will be formulated, where FMM, FSM, FTLM, FIM, and FAM will replace soybean at 0% (SBM100), 15% (FMM15, FSM15, FTLM15, FIM15, FAM15), 30% (FMM30, FSM30, FTLM30, FIM30, FAM30), and 60% (FMM60, FSM60, FTLM60, FIM60, FAM60). All diets will be processed into 2.0 mm pellets. All formulated feeds will meet the nutrient requirements of Nile tilapia and African Catfish as mentioned in da Silva et al. (2017) and Aliu & Olomu (2020), respectively. The chemical composition of FPM experimental diets, fish carcasses, and faecal matter will be analyzed according to AOAC procedures (1995). Crude protein will be determined using the micro-Kjeldahl method, crude lipid will be Soxhlet extraction with petroleum ether, and Nitrogen-free extract (NFE) content will be calculated using the Weende method. The anti-nutritional factors (phytic acid, tannins, trypsin inhibitor, hemagglutinin, aflatoxin, oxalate) will all be determined following Inuwa et al. (2011). Amino acid content will be determined using high-performance liquid chromatography (HPLC).

A total of 240 juvenile Nile Tilapia with initial weight (15.00 ± 0.20 g) will be randomly distributed into 20, 120-L plastic tanks, following a stocking density of 12 juveniles/tank with five (5) replicates for each formulated diet. Meanwhile, 300 juvenile African Catfish (20.00 ± 0.30 g) will be randomly stocked into the same number of tanks, following the same number of replications yet with a stocking density of 15 juveniles/tank. All the treatment diets will then be randomly assigned to each of the 20 tanks. This setup will be repeated 5 times until all formulated diets will be tested for the cultured species. In total, a total of 1,200 juvenile Nile Tilapia will be tested and a total of 1,500 Juvenile African Catfish will be purchased and be utilized for the whole experiment. Fish will be fed in three daily meals (08:00, 12:30, and 17:00 h). Water quality parameters for each tank will be monitored each week which includes water transparency, salinity, temperature, pH, dissolved oxygen (DO), and ammonia (da Silva et al. 2017) and will be measured using descriptive statistics (e.g., mean, SD). The growth performance and feed efficiency of the cultured fish will be evaluated using survival rate, weight gain, feed intake, feed conversion ratio, and protein efficiency ratio. Cultured species will be sampled and evaluated on the day before stocking and every two weeks throughout the culture period of 45 days to reduce stress on the species. The following formula will be used in this analysis.

$$\text{Survival rate (SR, \%)} = \left(\frac{\text{Final Fish Number}}{\text{Initial Fish Number}} \right) \times 100$$

$$\text{Weight Gain (WG, \%)} = \frac{\text{final body weight (g)} - \text{intial body weight (g)}}{\text{initial body weight (g)}} \times 100$$

$$\text{Feed Intake (FI)} = \left(\frac{\text{Total Feed Consumed (g)}}{\text{Number of fish per replicate}} \right)$$

$$\text{Feed Conversion Ratio (FCR)} = \left(\frac{\text{Total Feed Consumed (g)}}{\text{Weight Gain}} \right)$$

$$\text{Protein Efficiency Ratio (PER)} = 100 \times \left(\frac{\text{Weight Gain}}{\text{Protein Intake}} \right), \text{ with PI, } \frac{\text{g}}{\text{kg}} \text{ dry weight basis}$$

Statistical Analysis

Analysis of Variance (P<0.05) will be used to compare the ADC and nutrient composition of the

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formulated diet of FMM, FSM, FTLM, FIM, and FAM. Likewise, the same analysis will be used to compare the difference on the growth performance, feeding efficiency, and survival of the species of interest between the treatments of each of the formulated fermented plant-based protein diet and the differences among the five formulated plant-based protein diets: FMM, FSM, FTLM, FIM, FAM. Appropriate post-hoc analysis will be performed if necessary. Statistical Analysis will be done using Statistical Package for Social Sciences (SPSS) software (Version 22.0 for Windows; SPSS Inc., Chicago, IL, USA).

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

The timetable presented below only covers the experimentation for Phase 1.

Research Objectives	Year 1						Year 2					
	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
1. Preparation, procurement, and fermentation of Munggo Meal, Sitaw Meal, <i>Trichangthera</i> meal, <i>Indigofera</i> meal, and <i>Azolla</i> meal												
2. Digestibility Assay and Nutrient Composition Analysis												
3. Growth and Survival Performance Analysis												
4. Analysis of data gathered from observation												
5. Lay-outing, Editing and Printing												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible.)

Publication

- at least 6 articles on the potentiality of all the fermented plant-based protein sources; and
- at least 1 IEC material on feed formulation.

Patent/Intellectual Property

- at least 1 copyright filed for IEC material; and
- at least 6 copyrights filed for the articles.

Product

- 15 formulated fermented diets from 5 fermented plant-based protein sources (FMM, FSM, FTLM, FIM, FAM) and 1 reference diet

People Service

- at least two (2) young faculty and research personnel trained;

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- at least five (5) BSF student trained and graduated; and,
- Prior Art Search reports.

Place and Partnership

- Forged 1 partnership for each of the munggo and sitaw farmer, *Trichanthera*, *Indigofera*, and *Azolla* growers.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Initial results from the formulated diets of this proposal will be further optimized to determine the ideal formulation for the transfer of developed products to beneficiaries. Training on fermentation process and feed formulation will also be conducted to capacitate 2 young faculty and research personnel, 5 BSF students, and other interested individuals and beneficiaries.

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The target beneficiaries of this project are the key players in the aquafeed industry and the aquaculture growers and operators.

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Robert Christian F. Patani	Project Leader	Fisheries; Aquaculture
Gelene Roda I. Alvarez	Project Member	Fisheries; Aquaculture
Dr. Rey J. dela Calzada	Project Member	Fisheries; Aquaculture
Ianthe Marie P. Benliro	Project Member	Fisheries; Marine Fisheries
Alessandra Nicole B. Falalimpa	Project Member	Fisheries; Post-Harvest
Noreen Beah M. Alar	Project Member	Fisheries; Post-Harvest
Kate Alyssa G. Jore	Project Member	Fisheries; Aquaculture
Meldy Dorado	Project Member	Computer Science
TBI	Research Assistant	Fisheries; Aquaculture

Percent Time Devoted to the Program/ Project/ Study

Program/Project Leader (in hours): 3 hours/day

Program/Project Members (in hours): 1 hour/day

(18) Budgetary Requirement

The budgetary requirement presented below covers the experimentation for the Phase 1.

SOURCES OF EXPENSES	Quantity	Amount (PhP) / unit	Amount
A. Personnel Services (For studies needing Research Assistants, Informant/s, Guide)	1 RA for 3 months	12,000.00/mo	36,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)			
- Laboratory Supplies (Fish Samples, Supplies and Materials, Chromic Oxide, etc)			14,875.00
- Office Supplies (Record book, ballpen, marker, label, etc)			2,500.00
- Professional fees (laboratory analysis and other related fees)	4 diets	12,800.00	51,200.00
- Raw Materials and Feed Ingredients (Peanut, Munggo,			41,545.00

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Sitaw, Soybean, Fishmeal, Wheat, Gluten, AA, etc)			
- Miscellaneous (for incidental expenses)			10,000.00
Subtotal			120,120.00
C. CAPITAL OUTLAY			
- Aerator System (Air Pump/Compressor)	2 pcs	2,500.00	5,000.00
- 150-L Rectangular Tanks (Megabox) with fittings and pipings	32 pcs	1,500.00	48,000.00
- Mixer Grinder	1 pc	4,500.00	4,500.00
- Pelletizer	1 pc	40,000.00	40,000.00
Subtotal			97,500.00
Total			253,620.00

(19) Literature Cited (Include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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(20) Capsule Curriculum Vitae (One-page brief C.V. only)